MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, CAST, 45-DEGREE E-PLANE BEND, FULL CORRAL AND SELF JIGGING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

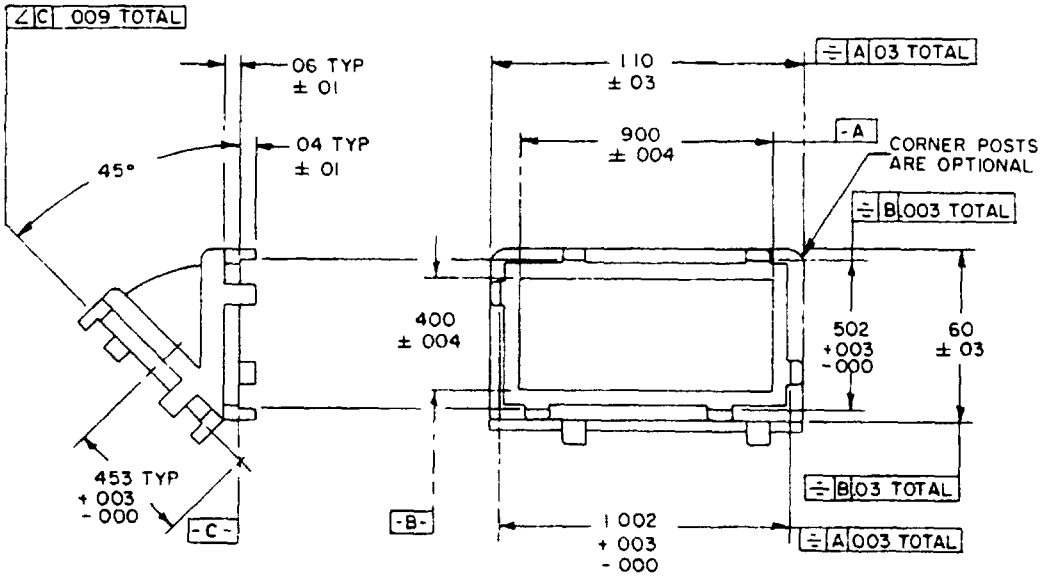
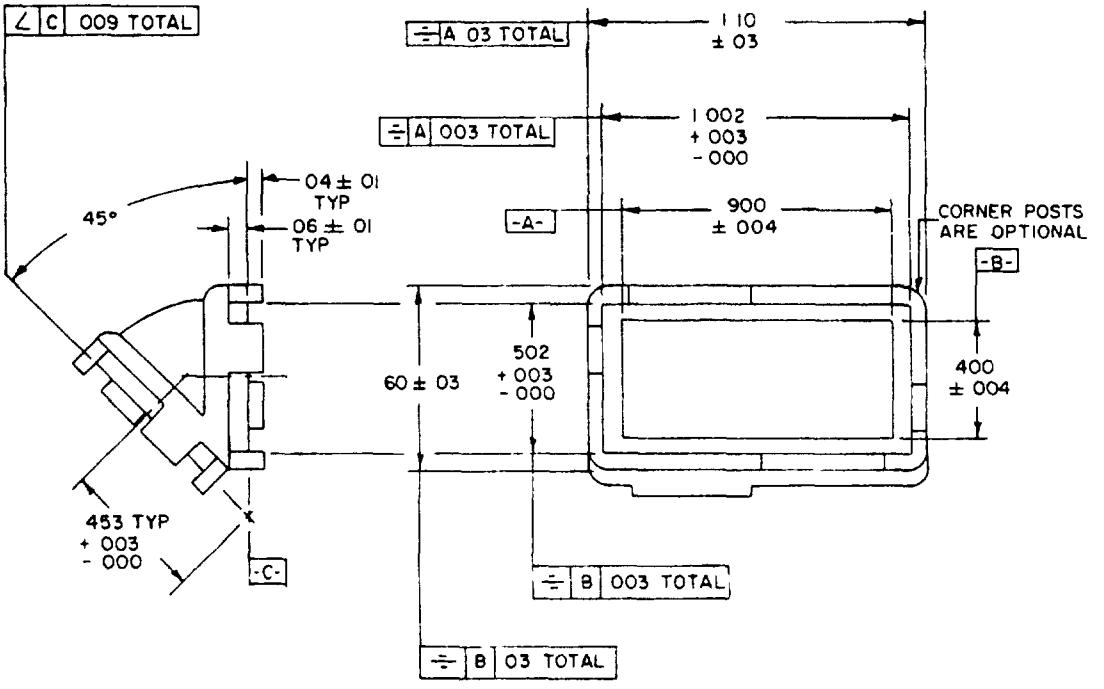


FIGURE 1 Dimensions for self-jigging waveguide assembly, for dash number 001

FSC 5985

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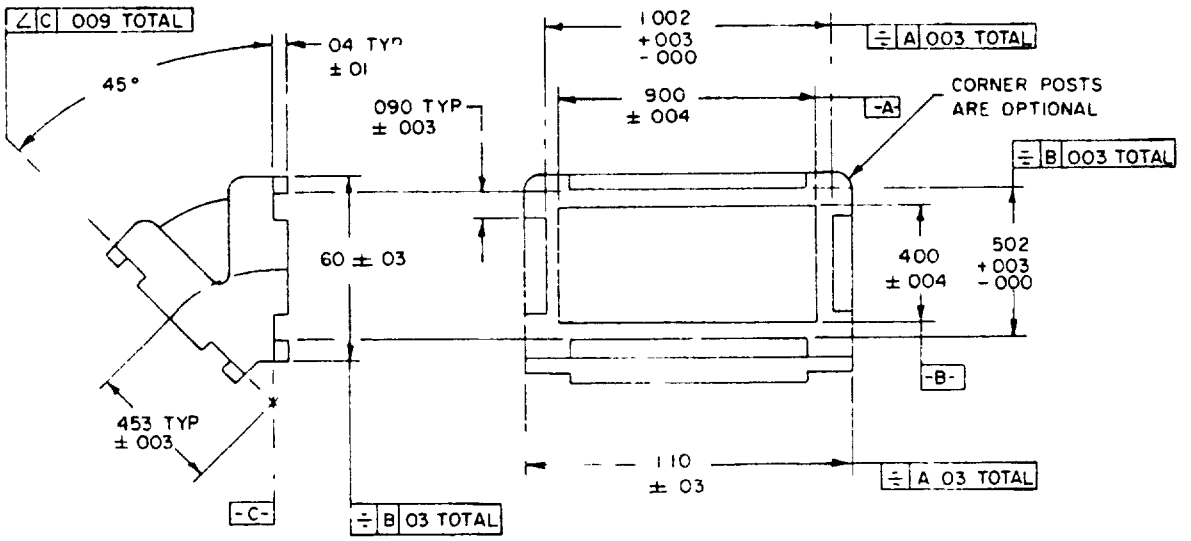
OPTIONAL CONFIGURATION

INCHES	MM	INCHES	MM
003	08	400	10 16
004	10	453	11 51
009	23	502	12 75
01	3	60	15 2
03	8	900	22 86
04	1 0	1.002	25 45
06	1 5	1 10	27 9

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1 Dimensions for self-jigging waveguide assembly, for dash number 001 - Continued

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INCHES	MM	INCHES	MM	INCHES	MM
003	08	03	8	502	12 75
004	10	04	1 0	60	15 2
009	23	090	2 29	900	22 86
01	3	400	10 16	1 002	25 45
		453	11 51	1 10	27 9

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 2 Dimensions for full corral configuration, for dash number 002

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REQUIREMENTS:

Physical characteristics
Dimensions and configuration See figures 1 and 2.
Operating temperature range -55° to +300°C.
Body material: D712.0, temper F, aluminum alloy per QQ-A-601 or C712.1 per QQ-A-371.
Surfaces (sandblasting after machining permissible)
Internal 63 $\sqrt{}$ microinches per ANSI B46.1.
External 125 $\sqrt{}$ microinches per ANSI B46.1.

Fungus resistance Materials nutrient to fungus shall not be used.

Pressurization 45 lb_f/in^2 gage maximum.

Penetrant inspection No indication of any cracks at any location when tested per MIL-I-6866, type I, method B.

Electrical characteristics
Frequency range 8.2 to 12.4 GHz.
VSWR 1.08 1 maximum.
Power
Peak - 250 kW.
Average - 10 kW.

Part number M3970/21-(dash number from figures 1 or 2).

NOTE The mating waveguides are M85/1-077, -078, and -178.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5985-0881-1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-W-3970/21

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFIC ENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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